

BLOCKCHAIN-BASED SECURE TRANSACTION SYSTEM USING JAVA

(Mini Project)

✧ Introduction :

Blockchain is a technology used to store and manage data in a secure and tamper-resistant manner. It stores information in a sequence of blocks, where each block is connected to the previous block using a cryptographic hash. The purpose of this mini project is to understand the fundamental concepts of blockchain by implementing a simple blockchain-based transaction system using Java.

In this project, transactions are stored inside blocks. Each block contains transaction information, a timestamp, its own SHA-256 hash, and the hash of the previous block. This creates a chain of connected blocks. The system also provides a blockchain verification feature. If information inside a block is modified, its hash changes, allowing the system to detect that the blockchain has been tampered with. This project is a simplified blockchain simulation intended for educational purposes. It demonstrates the core principles of blockchain without requiring external blockchain platforms.

✧ Problem Statement :

Traditional digital transaction records are generally maintained using centralized systems. If the stored information is modified without authorization, detecting such changes can be difficult.

The problem addressed by this project is to develop a simple system that demonstrates how blockchain concepts can be used to maintain transaction records and detect unauthorized modification of stored information.

The project implements a basic blockchain using Java in which:

- Transactions are stored in blocks.
- Each block contains a cryptographic hash.
- Each block stores the hash of the previous block.
- Blockchain integrity can be verified.
- Modification of block data can be detected.

✧ Objectives :

The main objectives of this project are:

1. To understand the basic concept of blockchain technology.
2. To implement a simple blockchain using Java.
3. To create blocks containing transaction information.
4. To generate SHA-256 cryptographic hashes.
5. To connect blocks using previous block hashes.
6. To implement blockchain verification.
7. To detect unauthorized modification of blockchain data.
8. To understand the relationship between transactions, blocks, and hashes.
9. To demonstrate the practical working of basic blockchain principles.

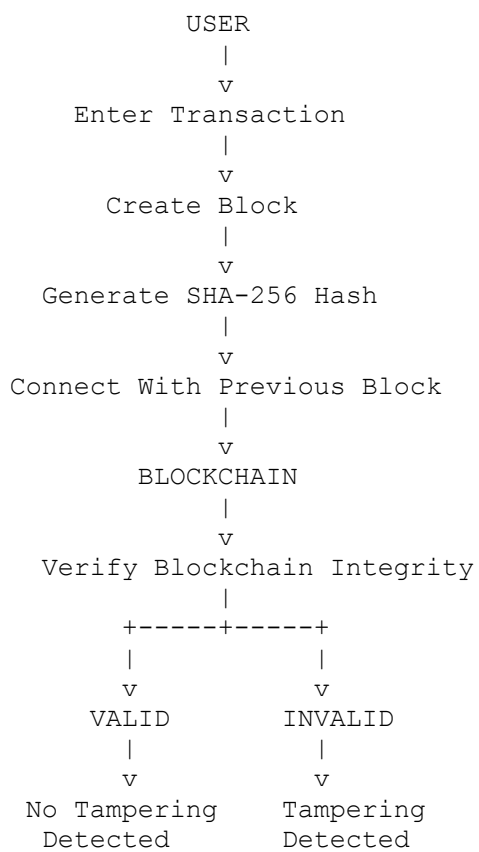
✧ Blockchain Platform and Tools Used :

The project uses a minimal technology stack to keep the implementation simple and easy to understand.

1. Java
2. Java SHA-256 API
3. Java IDE / VS Code

✧ System Architecture :

The overall architecture of the proposed system is:



❖ Screenshots and Outputs :

```
=====
BLOCKCHAIN SYSTEM
=====
1. Add Transaction
2. Display Blockchain
3. Verify Blockchain
4. Exit

Enter your choice: 1
Enter transaction: 2000

Block added successfully!
Block Number: 1
Hash: c5f1aef21cb76ede7514044686373dde68e9a97f7e28827a95041b9b9e043086

=====
BLOCKCHAIN SYSTEM
=====
1. Add Transaction
2. Display Blockchain
3. Verify Blockchain
```

```
===== BLOCKCHAIN =====

Block Number : 0
Transaction : Genesis Block
Previous Hash: 0
Hash       : 1f9a053df6f7e16da63397fa8d2e0bb8b692ae703e78dec0227b09154302cf8a
Timestamp  : 1790794758714

Block Number : 1
Transaction : 2000
Previous Hash: 1f9a053df6f7e16da63397fa8d2e0bb8b692ae703e78dec0227b09154302cf8a
Hash       : c5f1aef21cb76ede7514044686373dde68e9a97f7e28827a95041b9b9e043086
Timestamp  : 1790794766014

Block Number : 2
Transaction : 3000
Previous Hash: c5f1aef21cb76ede7514044686373dde68e9a97f7e28827a95041b9b9e043086
Hash       : 19d7a8efb4af0d2f93cf21f03c7510b9c77b0b65dd6f6ccb02977dcc1b601643
Timestamp  : 1790794799482
```

```
=====
BLOCKCHAIN SYSTEM
=====
1. Add Transaction
2. Display Blockchain
3. Verify Blockchain
4. Exit

Enter your choice: 3

Blockchain is VALID ?
No block has been tampered with.

=====
BLOCKCHAIN SYSTEM
=====
1. Add Transaction
2. Display Blockchain
3. Verify Blockchain
4. Exit
```

✧ Conclusion :

The project successfully demonstrates the fundamental concepts of blockchain technology using Java. Transactions are stored inside blocks, and every block contains a cryptographic hash and the hash of the previous block. This creates a chain of interconnected blocks. The SHA-256 hashing mechanism provides a way to detect modifications to block data. The blockchain verification feature checks both the block hash and the connection between consecutive blocks.

Through this project, the basic concepts of blocks, transactions, hashing, blockchain structure, and tamper detection can be understood through a practical implementation. The project provides a simple foundation for understanding more advanced blockchain technologies.